AWS Cloud Infrastructure

Name: Enoch Jason J
Class: AIML - A
Roll No: 24AM024
Date: 08/10/2026
Web page IP: 18.215.255.0

A "Hello World" page hosted on EC2 loads from the office network, but customers on mobile networks see "This site can't be reached." Propose at least five hypotheses, such as a missing inbound rule for port 80, NACL ephemeral port restrictions, a web server bound only to 127.0.0.1, a missing public IP, ISP-level port blocking, and users typing HTTPS when only HTTP is configured. Rank them by likelihood and ease of testing, and justify your order. Specify the tools you would use, including **curl -v, ss -tlnp, systemctl status httpd, and VPC Reachability Analyzer**. Explain why the public IP changes after a stop/start and how an **Elastic IP or a Route 53** record solves this. Finally, state where index.html must be placed (the document root) and which file permission or SELinux problems could cause a 403 error.

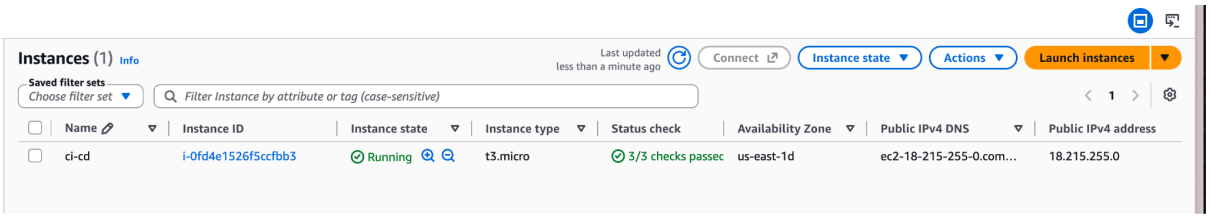


Fig 1. EC2 Dashboard

Select the Instance ID which will redirect you to the respective Instance's dashboard

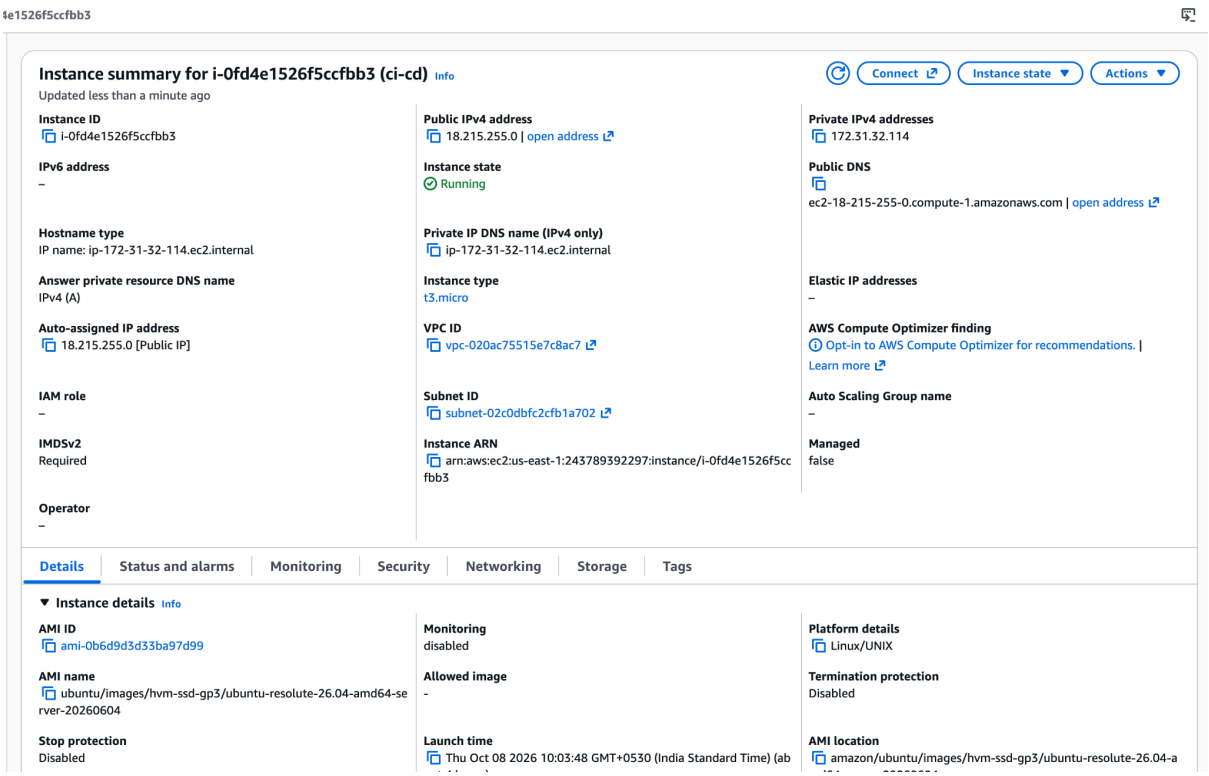


Fig 2. EC2 Instance dashboard

Click on the **Security** tab to edit the **Inbound** and **Outbound** configurations.

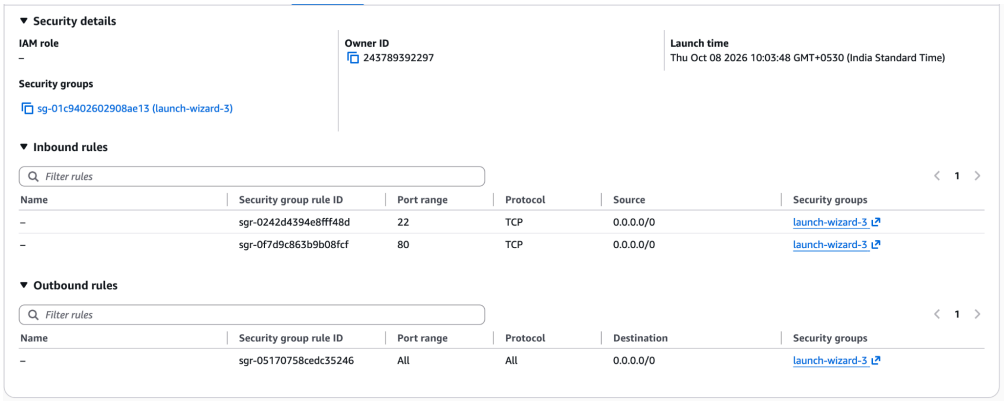


Fig 3. EC2 Security Group

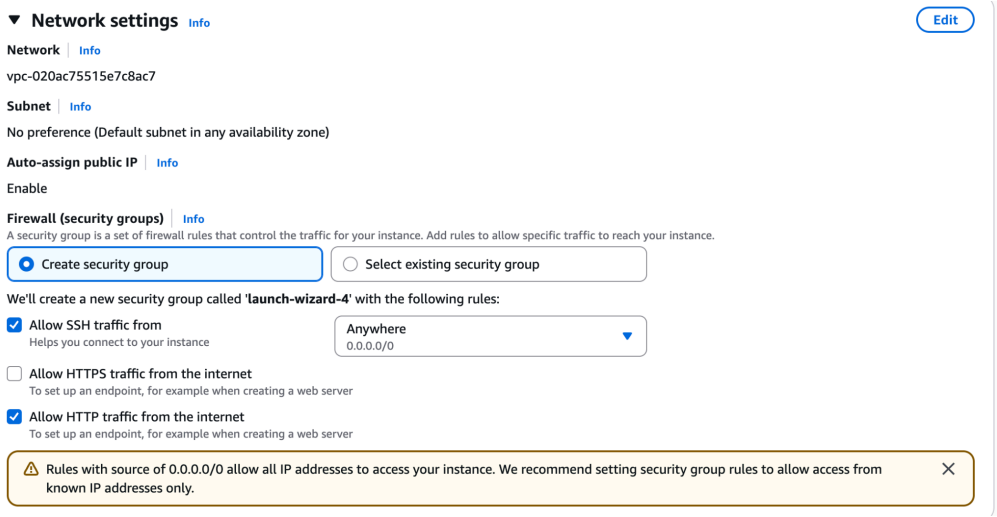


Fig 4. EC2 Network Settings

Enable the checkbox for **SSH traffic from** and **Allow HTTP traffic from the internet.** Then click **Launch Instance** button.

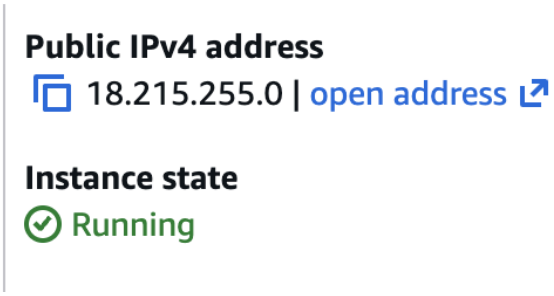


Fig 5. EC2 Instance public IP

Copy the **Public IPv4 address** of the instance.

deploy.yml × index.html .env

cicd > .github > workflows > deploy.yml

```yaml
name: CI/CD deploy to EC2

on:
  push:
    branches:
      - main

jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout repo code
        uses: actions/checkout@v3

      - name: Deploy to EC2
        uses: appleboy/scp-action@v0.1.7
        with:
          host: ${{secrets.EC2_HOST}}
          username: ${{secrets.EC2_USERNAME}}
          key: ${{secrets.EC2_SSH_KEY}}
          source: "."
          target: "/var/www/html"
```

Fig 6. Github Actions Workflow

Write the **Github Actions** configuration inside **.github/workflows/deploy.yml** along with the **index.html** file.

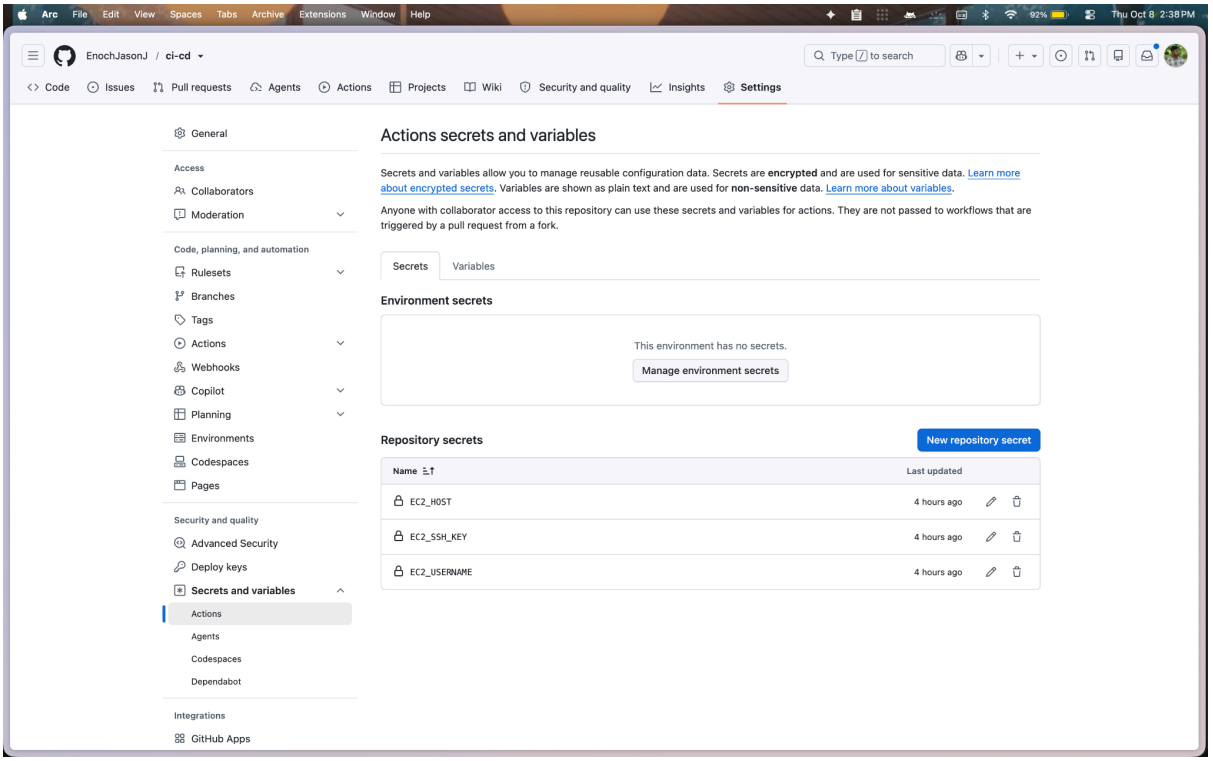


Fig 7. Github actions secrets & env variables

Configure the **AWS Credentials** inside **Repository->Settings-> Secrets and Variables -> Actions -> New repository secrets.**

```
❯ git add .
❯ git commit -m "Updated index.html file"
[main 74f4bd1] Updated index.html file
 2 files changed, 14 insertions(+), 2 deletions(-)
 create mode 100644 .DS_Store
❯ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 773 bytes | 773.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:EnochJasonJ/ci-cd.git
   80677ed..74f4bd1  main -> main

 /Volumes/Enoch/projects/AWS/s3/cicd  on git main !1
❯
```

Fig 8. Github commit

Push the contents to **github repository**

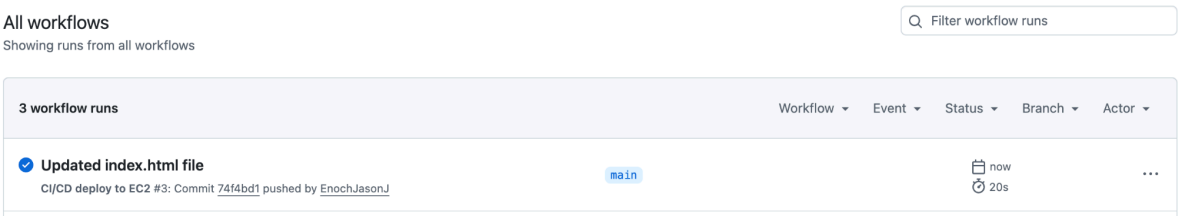
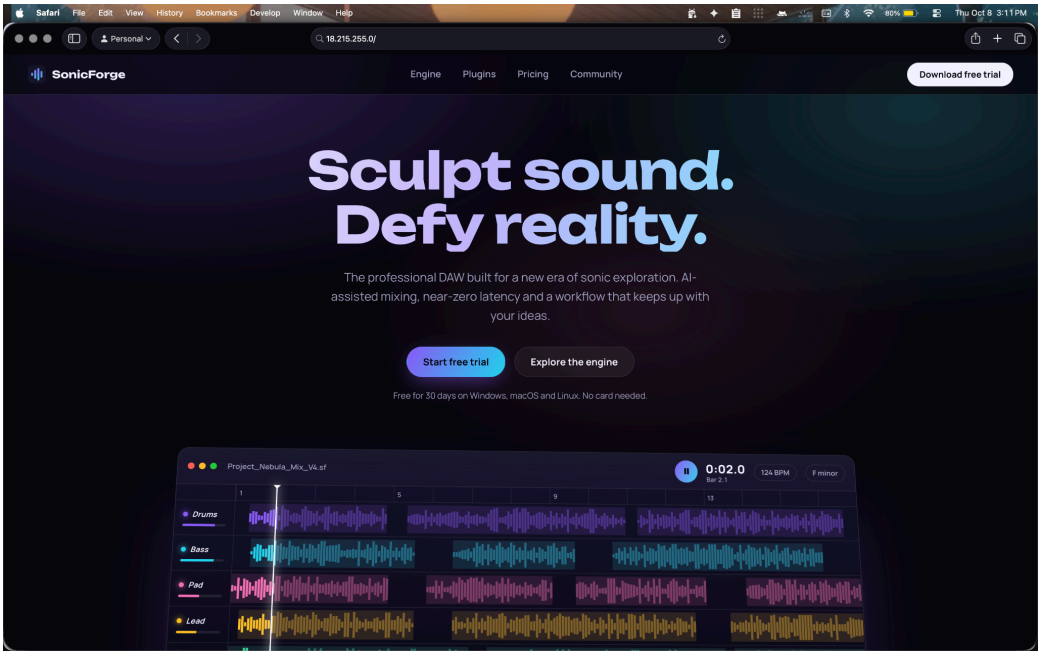
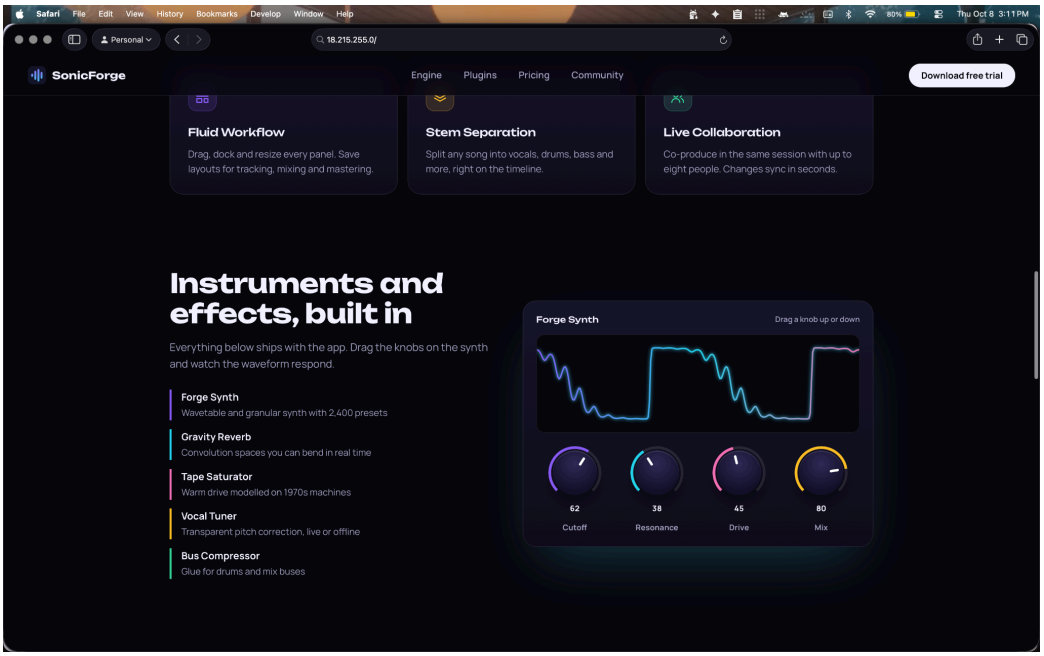


Fig 9. Github actions workflow run

Wait for the **Github Actions** to complete the deployment onto **AWS EC2**.

Open the deployed web page using the **Public IPv4 address.**